
EMG-Based Hand Gesture Classification

Ashwin Sakthivel

Problem & Dataset

Problem:

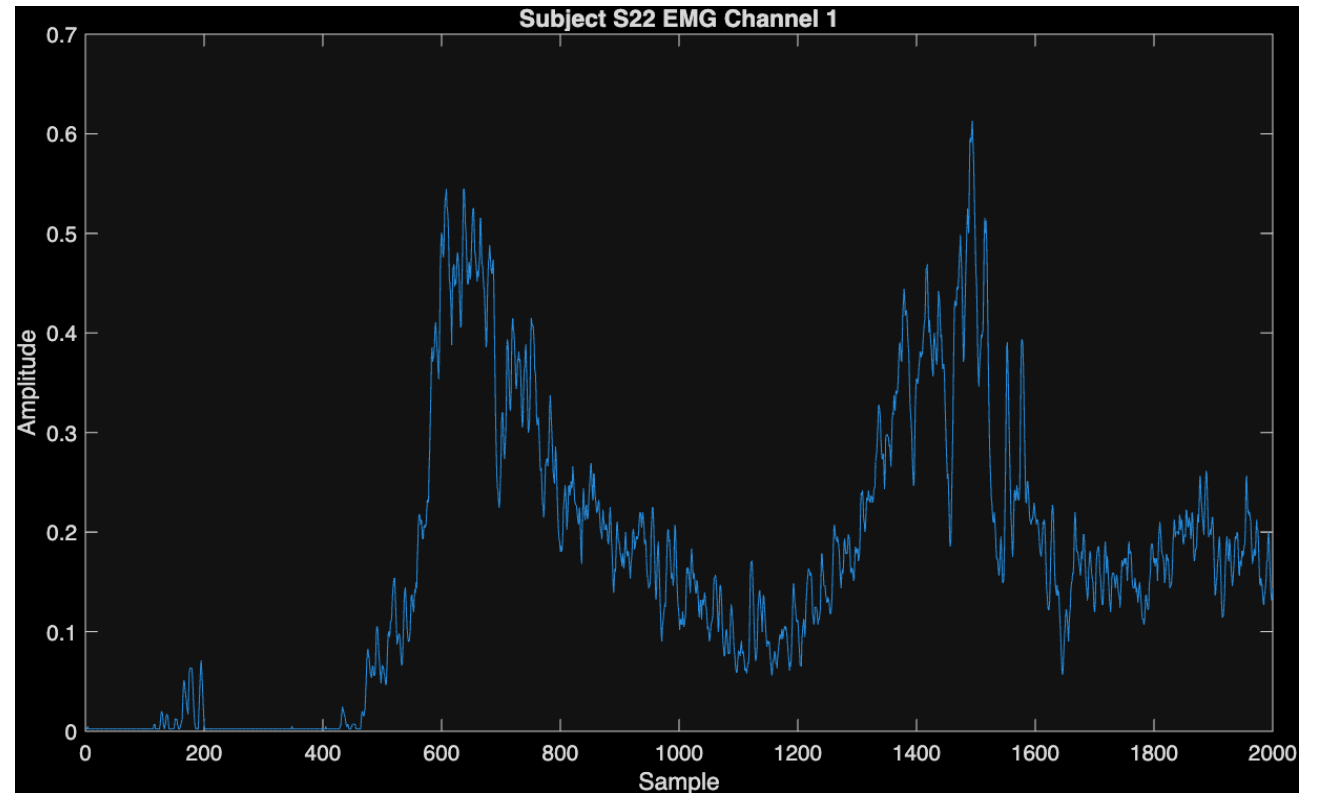
Recognize hand gestures using muscle activity measured from 10-channel sEMG.

NinaPro DB1, Exercise A1

23 gestures (we tested both 23-class & 12-class subsets)

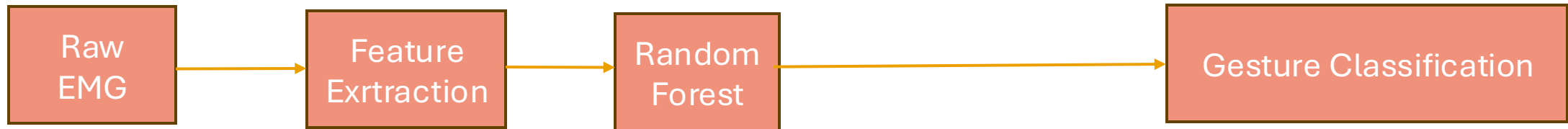
Each sample: 400-sample window $\times$ 10 channels

Preprocessed using 50 handcrafted features per window



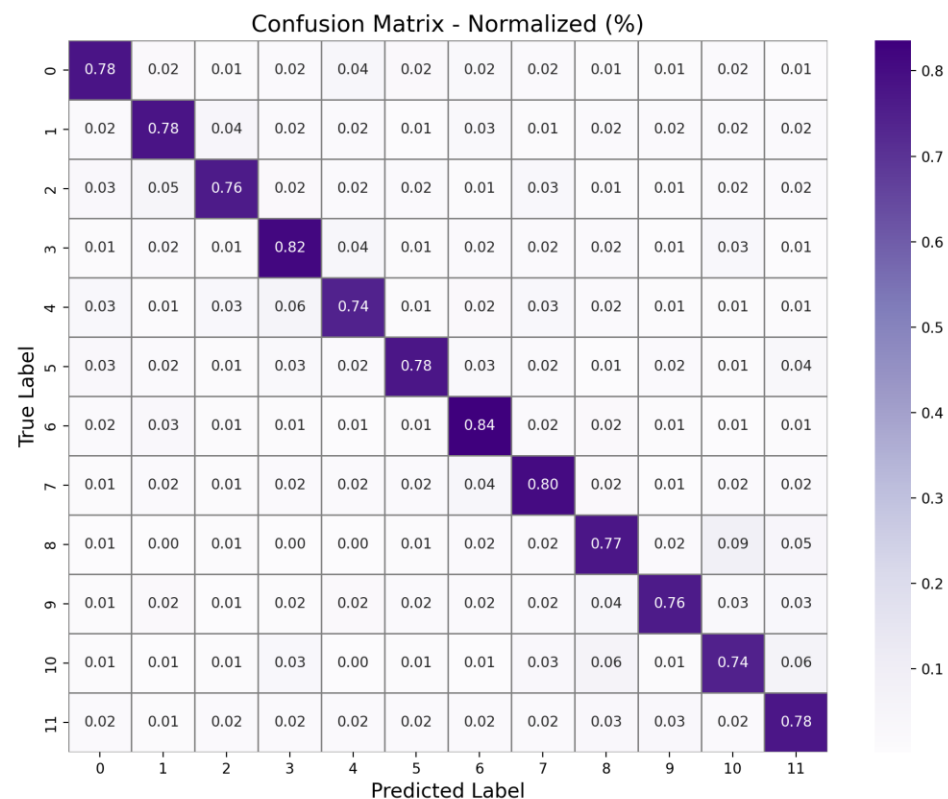
Method- Random Forest Pipeline

- Windowing (400 samples, 200-ms)
- EMG Features- Mean Absolute Value (MAV), Waveform Length (WL), Zero Crossings (ZC), Slope Sign Changes (SSC), RMS, Variance, Hjorth Parameters
- Random Forest Classifier 300 trees, max_depth = 30, Train/test = 80/20 split



Experiments & Setup

- Random Train/Test Results:
- Test Accuracy: 78.1%
- Strong performance across all 12–23 gesture classes
- Robust to noise due to ensemble averaging



Discussion, Insight, Demo

- **Insights:**

- Random Forests handled EMG variance well
- Handcrafted features remain extremely effective
- Lightweight model → real-time feasible
- Baseline outperformed deep-learning under limited preprocessing

- **Limitations:**

- Dataset contains high inter-subject variability
- Performance drops if using raw EMG instead of 50-dim features

